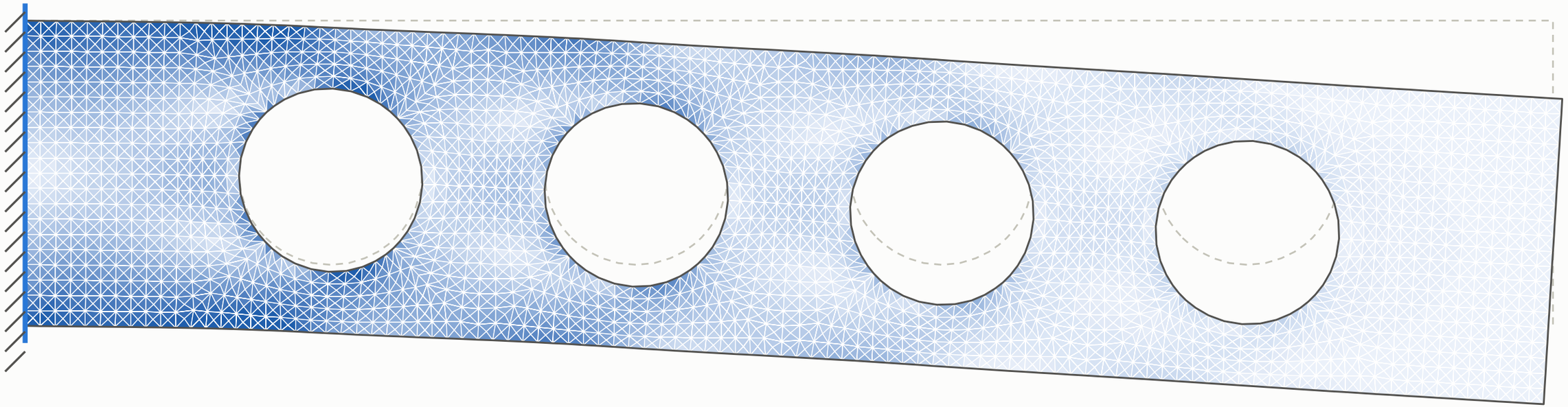
Minimum potential energy

Variational principles, the weak and the strong form, and linear elasticity in FEniCS



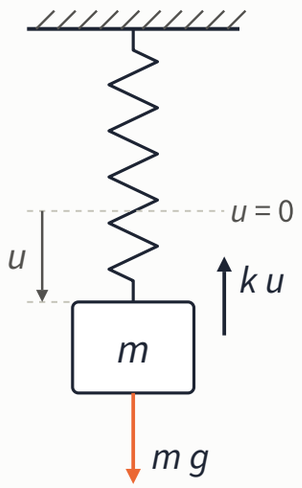
Variational principles in physics

Nature makes a functional stationary, often minimal. The physical law is the condition for it.

Physics regime		Variational principle	Strong form / physical law
Mechanics	Dynamics	**Stationary action** $\int \left(K - \Pi\right)\, dt$ stationary	**Newton's law** $m\,\frac{d^2x}{dt^2} = \sum F$
	Statics	**Minimum potential energy** THIS CHAPTER $\Pi = \Psi - W$ minimal	**Balance of forces** $\sum F = 0$
Optics		**Fermat's principle** travel time $\int \frac{n}{c}\, ds$ stationary	**Snell's law** $n_1 \sin\theta_1 = n_2 \sin\theta_2$
Electrostatics		**Dirichlet's principle** $\int \frac{1}{2}\,\epsilon\,\lvert\nabla\varphi\rvert^2 - \rho\,\varphi\, dx$ minimal	**Gauss's law** $-\nabla\cdot(\epsilon\,\nabla\varphi) = \rho$
Quantum mechanics		**Rayleigh–Ritz principle** $\langle\psi, H\psi\rangle \,/\, \langle\psi,\psi\rangle$ minimal	**Schrödinger equation** $H\psi = E\,\psi$

EXAMPLE

A mass on a spring



Balance of forces

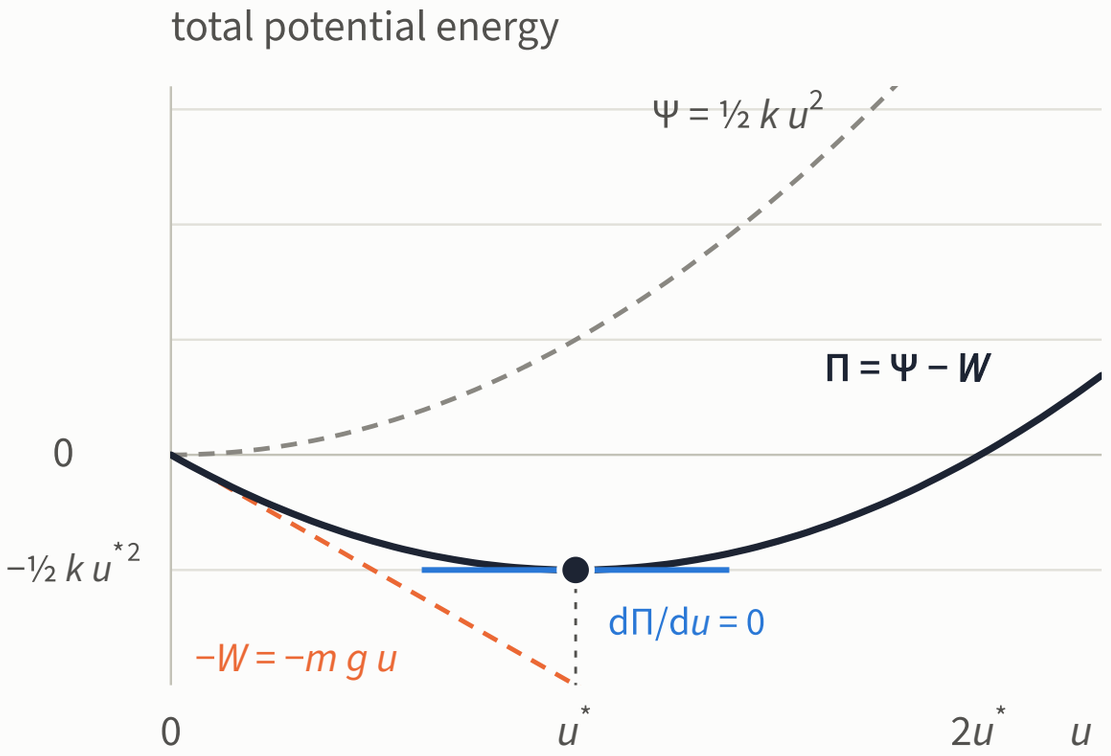
$$mg - ku = 0$$

hence $u^* = mg/k$

Minimum of the potential energy

$$\Pi(u) = \tfrac{1}{2}\,k\,u^2 - mg\,u$$

spring energy Ψ minus the work W of the weight



$d\Pi/du = ku - mg = -\sum F$: the minimum *is* the balance of forces.

The principle of minimum potential energy

A body in equilibrium takes the state u that **minimizes its total potential energy**

$$\Pi(u) = \Psi(u) - W(u)$$

among all **admissible** u: those that satisfy the essential boundary conditions.

Ψ: strain energy stored in the body, W: work of the applied loads

A symmetric, positive and linear

$$\Pi(u) = \tfrac{1}{2}\, a(u,u) - L(u)$$

with a and L of the variational formulation $a(u,v) = L(v)$.
Spring: $a(u,v) = k\,u\,v$, $L(v) = mg\,v$.

Why the minimizer solves $Au = f$

$$\Pi(v) = \tfrac{1}{2}\, a(v-u, v-u) - \tfrac{1}{2}\, a(u,u)$$

is smallest for $v = u$, with $\Pi(u) = -\tfrac{1}{2}\, a(u,u) = -\Psi(u)$

First variation and Euler–Lagrange equation

First variation

$$\delta\Pi(u;v) := \frac{d}{d\xi}\,\Pi(u+\xi v)\,\Big|_{\xi=0} = 0$$

Euler–Lagrange equation

For $\Pi(u) = \int_\Omega \mathcal{L}(x, u, \nabla u)\, dx$:

$$\frac{\partial \mathcal{L}}{\partial u} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla u} = 0$$

One functional, the weak and the strong form

FUNCTIONAL

$$\Pi(u) = \int_\Omega \mathcal{L}(x, u, \nabla u)\, dx - \int_\Gamma t\, u\, ds$$

energy in Ω, work of the load t on the Neumann boundary Γ

directional derivative

Euler–Lagrange equation (in Ω)

VARIATIONAL FORMULATION

$$\int_\Omega \frac{\partial \mathcal{L}}{\partial u}\, v + \frac{\partial \mathcal{L}}{\partial \nabla u} \cdot \nabla v\, dx - \int_\Gamma t\, v\, ds = 0$$

for all v

integrate by parts

STRONG FORM

$$\frac{\partial \mathcal{L}}{\partial u} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla u} = 0 \ \text{ in } \Omega$$

$$\frac{\partial \mathcal{L}}{\partial \nabla u} \cdot n = t \ \text{ on } \Gamma$$

on Γ: from integrating by parts

— Natural or Neumann

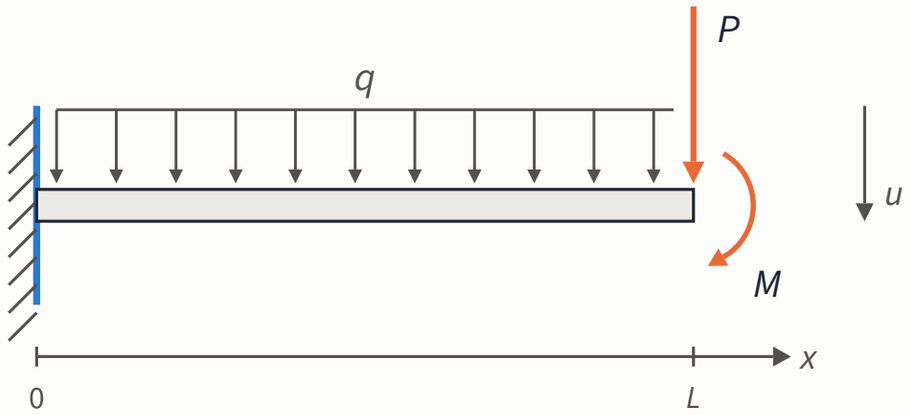
The boundary terms are part of Π: the natural conditions come out of the minimization.

● Essential or Dirichlet

Not in Π: they restrict the admissible u (`DirichletBC`).

EXAMPLE

The biharmonic equation: a beam



Clamped at $x = 0$: $u(0) = u'(0) = 0$.
Load q, end force P, end moment M.

$$\Pi(u) = \int_0^L \tfrac{1}{2}\, EI\, (u'')^2\, dx - \int_0^L q\, u\, dx - P\, u(L) - M\, u'(L)$$

Directional derivative, the weak form with $s = 2$:

$$\delta\Pi(u; v) = \int_0^L EI\, u''\, v''\, dx - \int_0^L q\, v\, dx - P\, v(L) - M\, v'(L) = 0$$

EXAMPLE

The beam: strong form and boundary conditions

Integrate by parts twice ($v(0) = v'(0) = 0$):

$$\int_0^L \left(EI\,u'''' - q\right) v\,dx + \left(EI\,u''(L) - M\right) v'(L) - \left(EI\,u'''(L) + P\right) v(L) = 0$$

$$EI\,u'''' = q \;\text{ in } (0, L)$$

	$x = 0$	$x = L$
essential	$u = 0,\ u' = 0$	–
natural	–	$EI\,u'' = M,\ -EI\,u''' = P$

Chapter 1, exercise 2

The load vector there is $F_j = L(h_j)$ with $L(v) = P\,v(1) + M\,v'(1)$:

$$(F_3, F_4) = (P, M) = (1, 0)$$

The boundary terms we dropped there are in Π.

The functional and the Galerkin method

Insert $u_h = \sum_i U_i \, h_i$:

Galerkin (chapter 1): test with $v = h_j$

$$a(u_h, h_j) = L(h_j)$$

Ritz: minimize Π over V_h

$$\Pi(U) = \tfrac{1}{2} U^T K U - U^T F$$

Both give $K U = F$: the Galerkin method *is* the minimization of Π.

- u_h is the **best approximation** in the energy norm.
- $\Pi(U)$ is the function that CG minimizes (chapter 2).

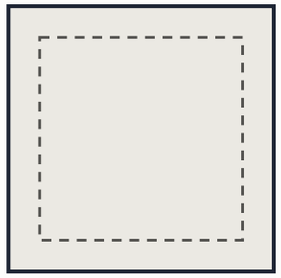
```
Pi  = (0.5*inner(grad(u), grad(u)) - f*u)*dx
dPi = derivative(Pi, u, v)     # delta Pi(u; v)
solve(dPi == 0, u, bc)
```

Displacement, strain and strain energy

The point x moves to $x + u(x)$; the **strain** is

$$\varepsilon(u) = \tfrac{1}{2}\left(\nabla u + \nabla u^T\right)$$

$\operatorname{tr}\varepsilon$: volume change, $\operatorname{dev}\varepsilon = \varepsilon - \tfrac{1}{3}(\operatorname{tr}\varepsilon)\, I$: change of shape



volume change
tr $\varepsilon \neq 0$, dev $\varepsilon = 0$
bulk modulus κ



change of shape
tr $\varepsilon = 0$, dev $\varepsilon \neq 0$
shear modulus G

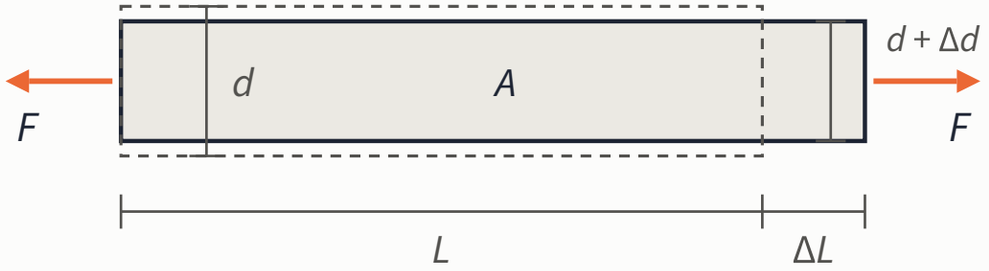
Strain energy density

$$\psi(\varepsilon) = \tfrac{1}{2}\,\kappa\,(\operatorname{tr}\varepsilon)^2 + G\,\operatorname{dev}\varepsilon : \operatorname{dev}\varepsilon$$

Stress

$$\sigma = \frac{\partial \psi}{\partial \varepsilon} = \kappa\,(\operatorname{tr}\varepsilon)\, I + 2G\,\operatorname{dev}\varepsilon$$

Measuring the constants



Tension test: the force F on the cross section A.

Young's modulus, Poisson's ratio

$$E = \frac{F/A}{\Delta L/L}, \qquad \nu = -\frac{\Delta d/d}{\Delta L/L}$$

Bulk and shear modulus (exercise 7):

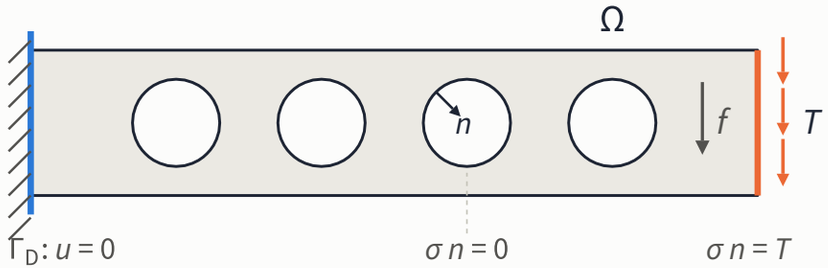
$$\kappa = \frac{E}{3\,(1-2\nu)}, \qquad G = \frac{E}{2\,(1+\nu)}$$

The examples use $E = 1000$ and $\nu = 0.3$.

The total potential energy of an elastic body

$$\Pi(u) = \int_\Omega \psi(\varepsilon(u))\, dx - \int_\Omega f \cdot u\, dx - \int_\Gamma T \cdot u\, ds$$

body force f, traction T on Γ, $u = 0$ on Γ_D



Directional derivative: the variational formulation of the dam (exercise 4)

$$\int_\Omega \sigma(u) : \varepsilon(v)\, dx = \int_\Omega f \cdot v\, dx + \int_\Gamma T \cdot v\, ds$$

Integrate by parts: the strong form

$$-\nabla \cdot \sigma(u) = f \text{ in } \Omega, \qquad \sigma n = T \text{ on } \Gamma$$

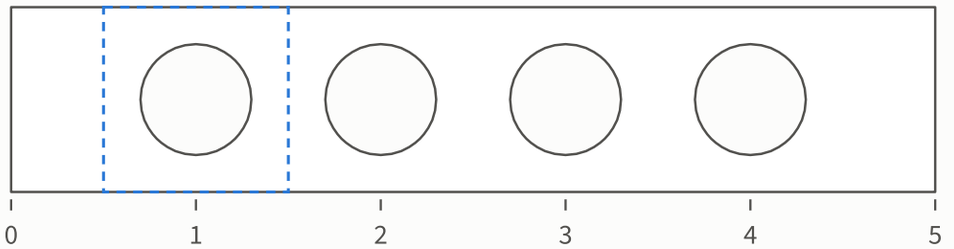
A beam with holes, without a mesh generator

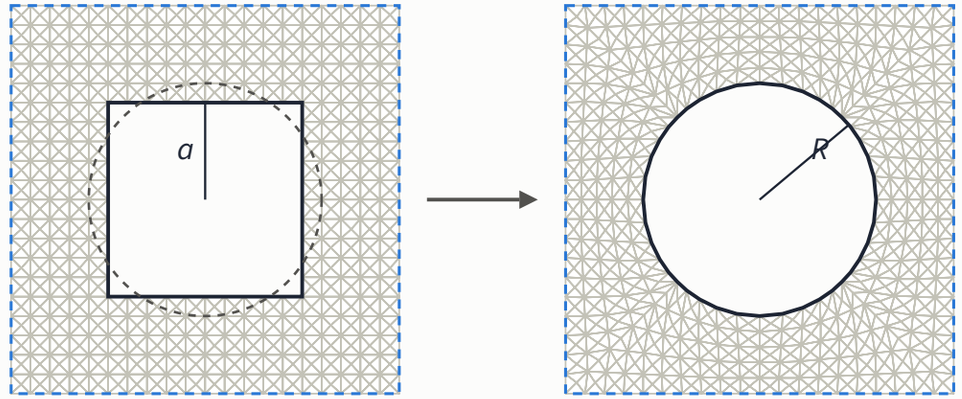
```python
from fenics import *
import numpy as np

length, height = 5.0, 1.0
R, a = 0.3, 0.25     # hole radius, half-width of the square cut out

box  = RectangleMesh(Point(0.0, 0.0), Point(length, height),
                     100, 20, 'crossed')
keep = MeshFunction('size_t', box, 2, 1)
for c in cells(box):     # cut out a square around each hole
    x, y = c.midpoint().x(), c.midpoint().y()
    if 0.5 < x < 4.5 and max(abs(x - round(x)), abs(y - 0.5)) < a:
        keep[c] = 0
mesh = SubMesh(box, keep, 1)

for X in mesh.coordinates():     # map the squares onto circles
    if 0.5 < X[0] < 4.5:
        centre = np.array([round(X[0]), 0.5])
        d = X - centre
        s, r = abs(d).max(), np.hypot(*d)
        t = (s - a)/(0.5 - a)    # 0 on the hole, 1 on the cell edge
        X[:] = centre + d*((1 - t)*R/r + t*0.5/s)
```





cut out: a square of half-width *a*

mapped: a circle of radius *R*

The square hole becomes a circle, the cell edges stay.
6400 triangles.

Shared setup: material, strain energy, boundaries

```
E, nu = 1000.0, 0.3
rho_g = 0.3
kappa = E/(3*(1 - 2*nu))    # bulk modulus
G     = E/(2*(1 + nu))      # shear modulus

V = VectorFunctionSpace(mesh, 'Lagrange', 1)

def epsilon(u):
    return sym(grad(u))

def psi(eps):               # strain energy density, plane strain
    return 0.5*kappa*tr(eps)**2 + G*(inner(eps, eps) - tr(eps)**2/3)

tol = 1e-3
def left(x, on_boundary):
    return on_boundary and near(x[0], 0.0, tol)

def right(x, on_boundary):
    return on_boundary and near(x[0], length, tol)

boundaries = MeshFunction('size_t', mesh, 1, 0)
AutoSubDomain(right).mark(boundaries, 1)       # ds(1): right end
ds = Measure('ds', domain=mesh, subdomain_data=boundaries)
```

Every example

1. write down `Pi`
2. `dPi = derivative(Pi, u, v)`
3. `solve(dPi == 0, u, bcs)`

Plane strain: $\varepsilon_{zz} = 0$, so `psi` takes the deviator in 3D.

Dirichlet – Dirichlet: a tension test

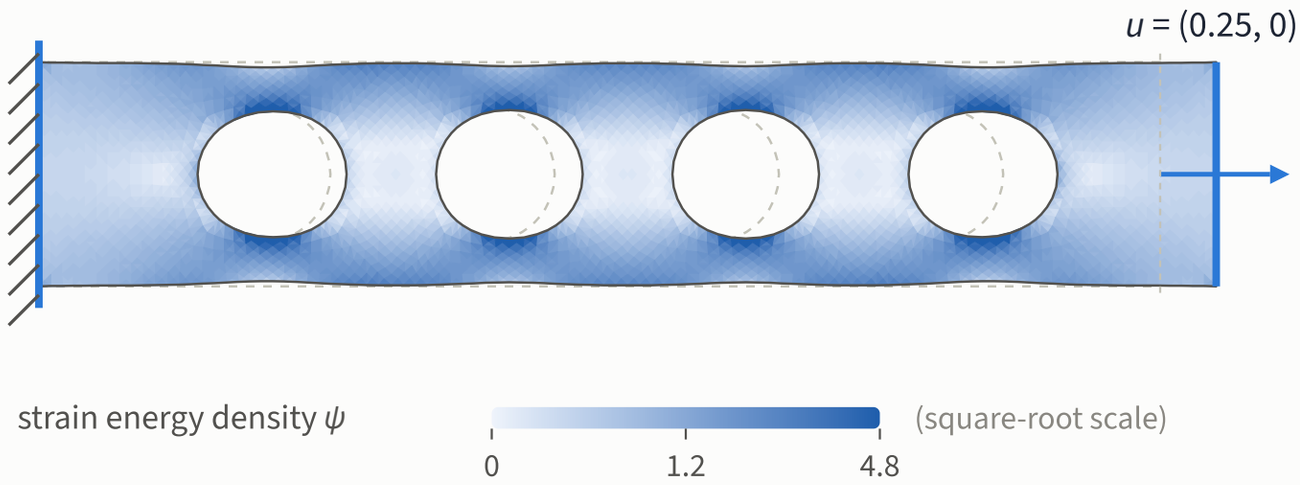
Left end fixed, right end pulled by 0.25; no loads: $\Pi(u) = \int_\Omega \psi(\varepsilon(u))\, dx$

```
bc0 = DirichletBC(V, Constant((0.0, 0.0)), left)
bc1 = DirichletBC(V, Constant((0.25, 0.0)), right)

u = Function(V)
v = TestFunction(V)

Pi  = psi(epsilon(u)) * dx
dPi = derivative(Pi, u, v)
solve(dPi == 0, u, [bc0, bc1])

print(assemble(Pi))     # stored strain energy
```



Stored energy 3.62, without the holes 6.94: the holes halve the stiffness.

Dirichlet – homogeneous Neumann: own weight

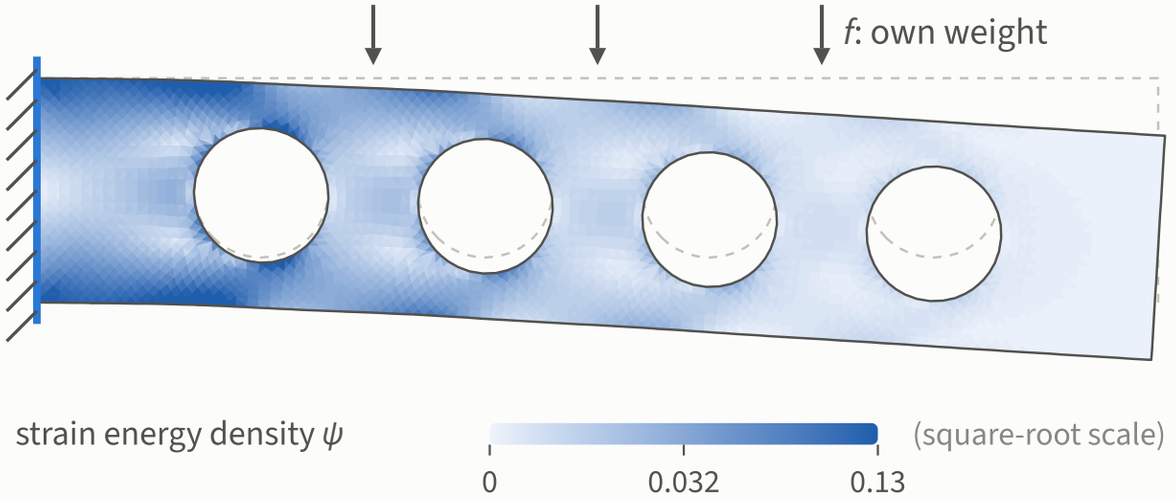
Left end clamped, body force $f = (0, -\rho g)$; no boundary term in Π.

```
bc = DirichletBC(V, Constant((0.0, 0.0)), left)
f  = Constant((0.0, -rho_g))

u = Function(V)
v = TestFunction(V)

Pi  = (psi(epsilon(u)) - dot(f, u)) * dx
dPi = derivative(Pi, u, v)
solve(dPi == 0, u, bc)

print(u(length, 0.5)[1])     # tip deflection
print(assemble(Pi), assemble(psi(epsilon(u))*dx))
```



Tip deflection -0.257, without the holes -0.265.

$\Pi(u) = -\Psi(u) = -0.064$, as for the spring.

- $\sigma n = 0$ on the holes and edges: no code.

Dirichlet – Neumann: an end load

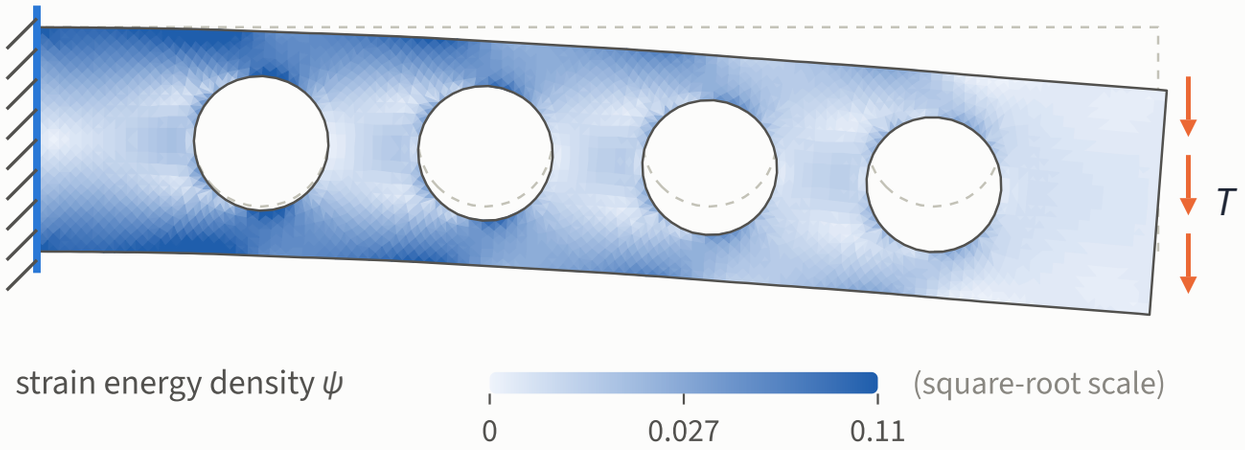
Left end clamped, traction $T = (0, -0.5)$ on the right end: the boundary term `ds(1)`.

```
T = Constant((0.0, -0.5))

u = Function(V)
v = TestFunction(V)

Pi  = psi(epsilon(u)) * dx - dot(T, u) * ds(1)
dPi = derivative(Pi, u, v)
solve(dPi == 0, u, bc)

print(u(length, 0.5)[1])     # tip deflection
print(assemble(Pi), assemble(psi(epsilon(u))*dx))
```



Tip deflection -0.282, without the holes -0.233: 21 % softer.
$\Pi(u) = -\Psi(u) = -0.071$.

Minimum potential energy in four steps

1. Write down $\Pi = \Psi - W$, with the work of all loads, also on the boundary.
2. $\delta\Pi(u; v) = 0$ for all v: the variational formulation (`derivative`).
3. Integrate by parts: the strong form and the **natural boundary conditions**.
4. Minimize over V_h: $K\,U = F$, the Galerkin method (`solve(dPi == 0, u, bcs)`).

EXAMPLE

A suspension bridge under its own weight

Dimensions

- main span 12, two side spans 4: length 21
- deck: width 2, thickness 0.3
- towers 3.35 above the deck, piers 2.2 below it
- two cables, sag 2.7 in the main span, 2×23 hangers

Material and load

$E = 1000,\ \nu = 0.3$, own weight $f = (0, 0, -\rho g)$ with $\rho g = 0.3$, as in the 2D examples

Variational form

Fixed ($u = 0$) at the feet of the piers and at the abutments: Γ_D.

$$\Pi(u) = \int_\Omega \psi(\varepsilon(u))\, dx - \int_\Omega f \cdot u\, dx$$

$\delta\Pi(u; v) = 0$: find u with $u = 0$ on Γ_D such that

$$\int_\Omega \sigma(u) : \varepsilon(v)\, dx = \int_\Omega f \cdot v\, dx$$

for all v with $v = 0$ on Γ_D; all other surfaces are free, $\sigma n = 0$

EXAMPLE

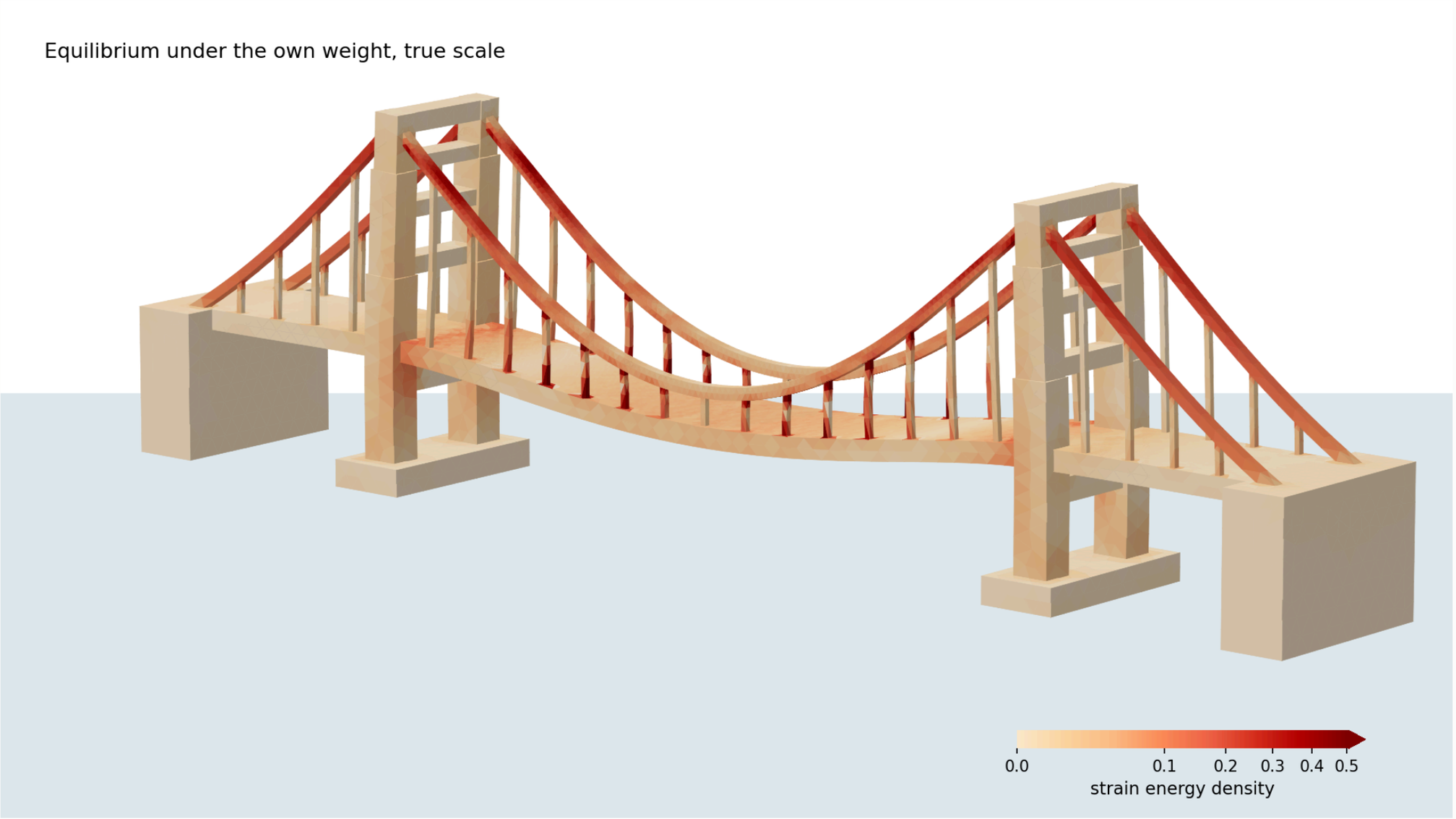
The mesh



16 426 nodes, 54 830 tetrahedra, 49 278 unknowns (P1). Built from boxes and two cables with Gmsh.

EXAMPLE

On the way to the minimum of Π



EXERCISE 1

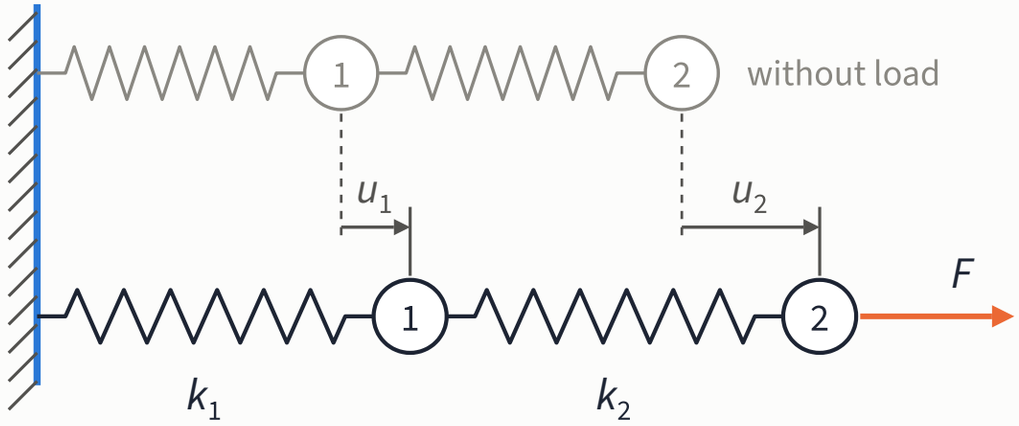
The free energy of stationary heat conduction

Problem of chapter 1: $-\nabla \cdot \nabla u = f$ in Ω, u prescribed on $\partial\Omega \setminus \Gamma$, outward heat flux $q \cdot n = g$ with $q = -\nabla u$ on Γ. Consider the free energy

$$F(u) = \int_\Omega \tfrac{1}{2} \left|\nabla u\right|^2 \, dx - \int_\Omega f\, u \, dx + \int_\Gamma g\, u \, ds.$$

1. Compute $\delta F(u; v)$ and compare with the variational formulation of chapter 1.
2. Apply the Euler–Lagrange equation to F.
3. Multiply the result of 2 by a test function v and integrate by parts. Compare with 1: which boundary condition holds on Γ, and why does the boundary term of F carry a plus sign?
4. Solve problem 3 of chapter 1 ($u(0) = 0$, $f = 0$, $g = -1$ at $x = 1$, 20 elements) as a minimization problem with `derivative` and compare with the solution from `a == L`.

Two springs

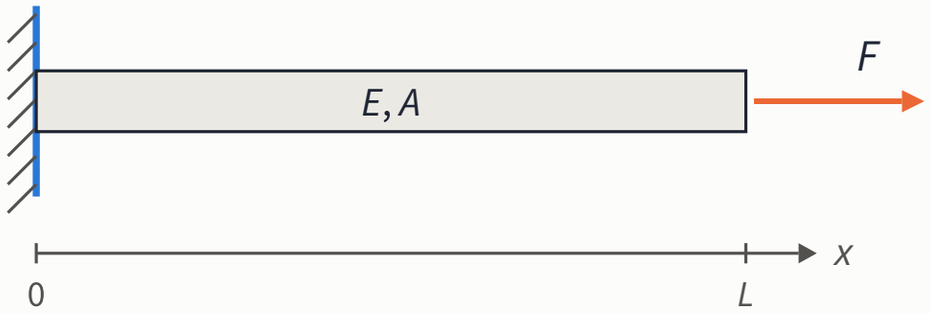


Two springs in a row, fixed to a wall and pulled by the force F. The nodes move by u_1 and u_2 from their positions without load:

$$\Pi(u_1, u_2) = \tfrac{1}{2}\, k_1 u_1^2 + \tfrac{1}{2}\, k_2 \,(u_2 - u_1)^2 - F\, u_2$$

1. Set $\partial\Pi/\partial u_1 = 0$ and $\partial\Pi/\partial u_2 = 0$, and write the two equations as $\underline{\underline{K}}\,\underline{U} = \underline{F}$.
2. Read each equation as a balance of forces at a node.
3. Show that the stationary point is a minimum.
4. Solve for u_2 in the case $k_1 = k_2 = k$.

A bar: Hooke's law in 1D



A bar of length L and cross section A, fixed at $x = 0$ and pulled by the force F at $x = L$. With the displacement $u(x)$: strain $\varepsilon = u'$, stress $\sigma = E\,\varepsilon$ (Hooke's law).

$$\Pi(u) = \int_0^L \tfrac{1}{2}\, EA\,(u')^2\, dx - F\,u(L)$$

1. Compute $\delta\Pi(u; v)$.
2. Apply the Euler–Lagrange equation to Π.
3. Integrate $\delta\Pi$ by parts. Which condition holds at $x = L$, and what does it say about the stress?
4. Solve for u and compute ε and σ. Compare with E in the tension test. Which spring stiffness k has the bar?

EXERCISE 4

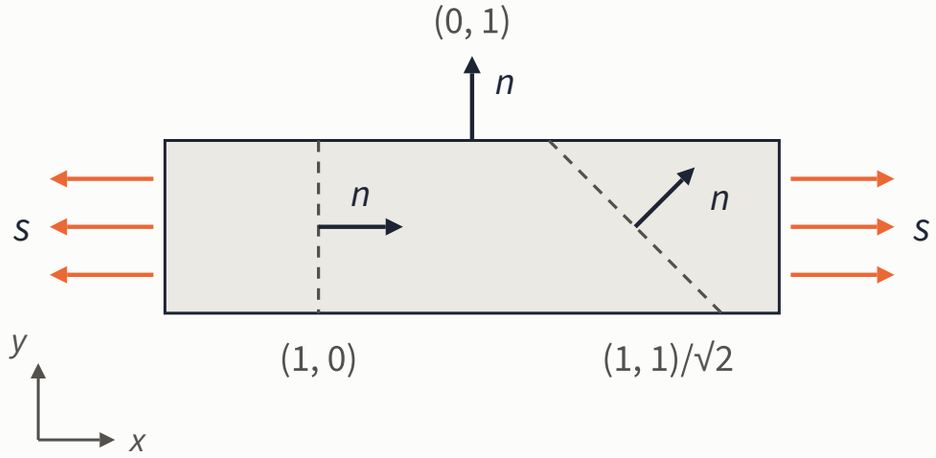
Stress: the force on a surface

The stress σ is a linear map: on a surface with the unit normal n, the force per unit area is σn.

1. Compute σn

   a. for a bar under tension, $\sigma = \begin{pmatrix} s & 0 \\ 0 & 0 \end{pmatrix}$, with $n = (1, 0), n = (0, 1)$ and $n = (1, 1)/\sqrt{2}$;

   b. for water at rest, $\sigma = -p\, I$ with the pressure p, and any n. In which direction does the water push on a wall?



Linear elasticity: a dam

Find u such that, for all v,

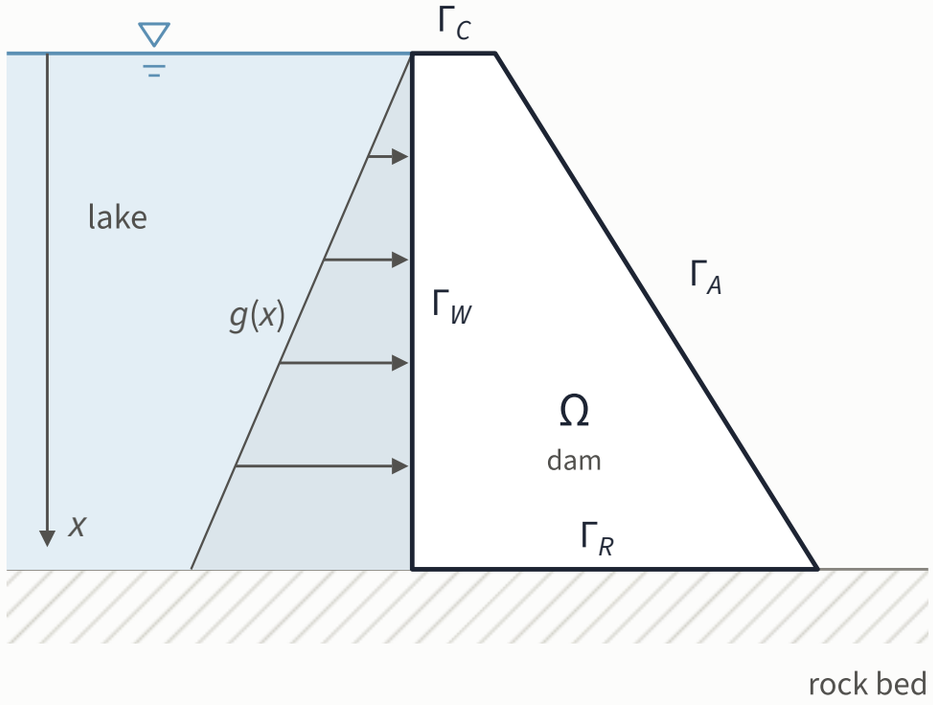
$$\int_\Omega \sigma(u) : \varepsilon(v)\, dx = \int_\Omega f \cdot v\, dx + \int_{\partial\Omega} T \cdot v\, ds,$$

$$\varepsilon(u) = \tfrac{1}{2}\left(\nabla u + \nabla u^T\right), \qquad \sigma(u) = \lambda\,(\operatorname{tr}\varepsilon)\, I + 2\mu\,\varepsilon.$$

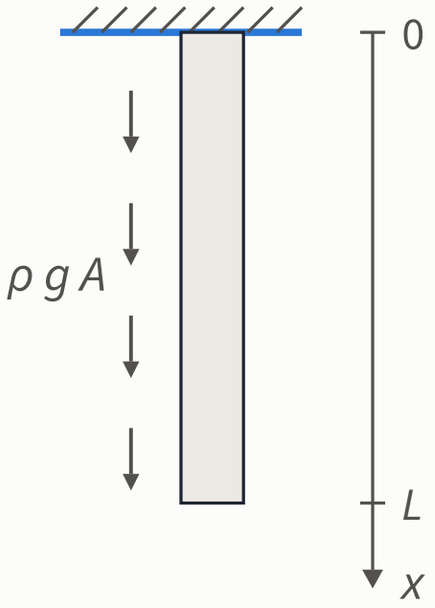
$T = \sigma n$: traction on the boundary; f: body force.

Hydrostatic pressure of the water at the depth x: $g(x) = \rho\, g\, x$.

2. Which of $\Gamma_W, \Gamma_R, \Gamma_C, \Gamma_A$ are Dirichlet, which Neumann boundaries?
3. Modify the variational form accordingly. Item 1 gives T on the water side.



Ritz by hand: a hanging bar



A bar hangs from the ceiling, $u(0) = 0$, and stretches under its own weight $\rho g A$ per unit length:

$$\Pi(u) = \int_0^L \tfrac{1}{2}\, EA\,(u')^2 - \rho g A\, u \, dx$$

1. Insert $u_h = c\,x$ and minimize Π over c.
2. Insert $u_h = c_1 x + c_2 x^2$ and minimize over c_1 and c_2: a system $\underline{\underline{K}}\,\underline{U} = \underline{F}$.
3. Compare the two minimal energies and the displacements of the lower end. Which u_h solves $-E\,u'' = \rho g$ with $u(0) = 0$ and $u'(L) = 0$?

A simply supported beam

A beam of length L and stiffness EI under the uniform load q, supported at both ends: $u(0) = u(L) = 0$, the ends can rotate.

1. Which boundary conditions are essential, which natural? Write down $\Pi(u)$ and the admissible functions.
2. Derive the strong form and the natural boundary conditions from $\delta\Pi(u; v) = 0$ by integrating by parts twice. What is the bending moment $EI\,u''$ at the supports?
3. Solve $EI\,u'''' = q$ with these boundary conditions. Give the deflection at midspan and the slope at $x = 0$.

Material constants (theory)

1. Show that ψ equals $\frac{1}{2}\,\lambda\,(\operatorname{tr}\varepsilon)^2 + \mu\,\varepsilon : \varepsilon$ with $\lambda = \kappa - \frac{2}{3}G$, $\mu = G$, and compute $\sigma = \partial\psi/\partial\varepsilon$.
2. In the tension test $\sigma = \mathbf{diag}(\sigma_{xx}, 0, 0)$. Show that $\varepsilon = \mathbf{diag}(\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{yy})$ with

$$E = \frac{\sigma_{xx}}{\varepsilon_{xx}} = \frac{9\,\kappa G}{3\kappa + G}, \qquad \nu = -\frac{\varepsilon_{yy}}{\varepsilon_{xx}} = \frac{3\kappa - 2G}{2\,(3\kappa + G)}.$$

3. Invert the relations to obtain κ, G and λ from E and ν, and compute the values for $E = 1000$, $\nu = 0.3$. What happens for $\nu \to \frac{1}{2}$?